

System Data Flow

Level 0

Sensors



Edge Gateway



Cloud Platform



Analytics Engine



Dashboard



Users

Data Sources

Source	Frequency
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Temperature	5 sec
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Source	Frequency
Vibration	5 sec
Humidity	10 sec
Energy Usage	30 sec

Processing Pipeline

Data Collection

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Validation

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Cleaning

↓

Storage

↓

Analytics

↓

Alert Generation

↓

Visualization

Output

- Live Dashboard
- AI Predictions
- Maintenance Alerts
- Reports

Storage

Raw Data



Processed Data



Analytics Dataset



Historical Archive